HDOC DAY – 42

ALGORITHM:

1. Start.
2. Display the menu:
    - Remove all vowels from a string.
    - Find the first repeating lowercase alphabet.
    - Check whether two strings are anagrams.
    - Exit.
3. Read the user's choice.
4. If choice is 1:
    - Read a string.
    - Scan each character.
    - Check whether the character is a, e, i, o, u or their uppercase forms.
    - If it is not a vowel, copy it to the resulting string.
    - Add the null character '\0' at the end.
    - Display the string without vowels.
5. If choice is 2:
    - Read a string.
    - Initialize a frequency array of size 26 with zero.
    - Scan each lowercase alphabet.
    - Increase its frequency.
    - When a character's frequency becomes 2 for the first time, display it as the first repeating lowercase alphabet.
    - If no character repeats, display an appropriate message.
6. If choice is 3:
    - Read two strings.
    - Compare their lengths.
    - If their lengths differ, they are not anagrams.

- Otherwise, create a frequency array.
- Increase frequency using characters of the first string.
- Decrease frequency using characters of the second string.
- If all frequencies become zero, the strings are anagrams.
- Otherwise, they are not anagrams.

7. If choice is 4, exit.
8. For any other choice, display an invalid-choice message.
9. Repeat until Exit is selected.
10. Stop.

C PROGRAM:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

// Function to remove all vowels
void removeVowels()
{
   char str[200];
   int i, j = 0;

   printf("Enter a string: ");
   getchar();
   fgets(str, sizeof(str), stdin);

   for (i = 0; str[i] != '\0'; i++)
   {
      if (str[i] != 'a' && str[i] != 'e' &&
         str[i] != 'i' && str[i] != 'o' &&
```

```
            str[i] != 'u' && str[i] != 'A' &&
            str[i] != 'E' && str[i] != 'I' &&
            str[i] != 'O' && str[i] != 'U')
        {
            str[j] = str[i];
            j++;
        }
    }

    str[j] = '\0';

    printf("String after removing vowels: %s", str);
}


// Function to find first repeating lowercase alphabet
void firstRepeating()
{
    char str[200];
    int freq[26] = {0};
    int i;

    printf("Enter a string: ");
    getchar();
    fgets(str, sizeof(str), stdin);

    for (i = 0; str[i] != '\0'; i++)
    {
```

```
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            freq[str[i] - 'a']++;

            if (freq[str[i] - 'a'] == 2)
            {
                printf("First repeating lowercase alphabet: %c\n", str[i]);
                return;
            }
        }
    }

    printf("No repeating lowercase alphabet found.\n");
}


// Function to check anagrams
void checkAnagram()
{
    char str1[100], str2[100];
    int freq[26] = {0};
    int i;

    printf("Enter first string: ");
    scanf("%99s", str1);

    printf("Enter second string: ");
    scanf("%99s", str2);
```

```
    if (strlen(str1) != strlen(str2))
    {
        printf("The strings are NOT anagrams.\n");
        return;
    }

    for (i = 0; str1[i] != '\0'; i++)
    {
        freq[tolower(str1[i]) - 'a']++;
        freq[tolower(str2[i]) - 'a']--;
    }

    for (i = 0; i < 26; i++)
    {
        if (freq[i] != 0)
        {
            printf("The strings are NOT anagrams.\n");
            return;
        }
    }

    printf("The strings are anagrams of each other.\n");
}


// Main function
int main()
```

```
{
    int choice;

    do
    {
        printf("\n----- MENU -----\n");
        printf("1. Remove all vowels from a string\n");
        printf("2. Find first repeating lowercase alphabet\n");
        printf("3. Check if two strings are anagrams\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                removeVowels();
                break;

            case 2:
                firstRepeating();
                break;

            case 3:
                checkAnagram();
                break;
```

```
        case 4:
            printf("Program terminated.\n");
            break;

        default:
            printf("Invalid choice! Please try again.\n");
    }

} while (choice != 4);

return 0;
}
```

TEST CASE TABLE:

Test Case	Choice	Input	Expected Output
1	1	education	dctn
2	1	Hello World	Hll Wrld
3	2	programming	First repeating lowercase alphabet: r
4	2	abcdef	No repeating lowercase alphabet
5	3	listen, silent	Strings are anagrams
6	3	earth, heart	Strings are anagrams
7	3	hello, world	Strings are NOT anagrams
8	4	—	Program terminated